

Part A Set X

Birla Institute of Technology & Science, Pilani
First Semester 2023-24
CS F111 – Computer Programming
Comprehensive Examination

Invigilator's Signature:

[Signature]

13/12/2023

Max. Marks: 45M

Duration: 75 mins

ID No:

Name:

Instructions:

- This is an in-built question paper. You need to tick (✓) the correct answers in this paper itself and return.
- Each question carries 1.5 mark without any negative marking. Only one option is correct for each question.
- Over-written answers of any kind will not be accepted for rechecks.
- You are not allowed to go outside the examination hall until you submit this question paper back.

Recheck requests	Remarks

1. Consider the declaration given below.

```
struct node {  
    char i;  
    double j; };  
struct node *s[10];
```

The above C declaration defines:

- ☒ A. An array, each element of which is a pointer to a structure of type node
 - ☐ B. A structure of 2 fields, each field being a pointer to an array of 10 elements
 - ☐ C. A structure of 3 fields: a character, a double, and an array of 10 elements
 - ☐ D. An array, each element of which is a structure of type node.
2. Choose a correct statement about C file "r" mode operation using fopen.
fopen("records.txt", "r");
- ☐ A. If the file records.txt is not found, fopen returns NULL or 0.
 - ☐ B. If the file records.txt is found, fopen returns a FILE pointer.
 - ☐ C. File records.txt is opened in Read Mode.
 - ☒ D. All the above
3. Given that the contents of a file named records.txt is: **CS F111 exam.**
Which of the options below indicate the correct output of the following code snippet?

```
char ch;  
FILE *fp;  
fp = fopen("records.txt", "w");  
while ((ch = fgetc(fp)) != EOF){  
    printf("%c", ch);}  
printf("Good Luck.");  
fclose(fp);
```

- ☐ A. CS F111 exam. Good Luck.
- ☒ B. Good Luck.
- ☐ C. CS F111 exam. and enters into an infinite loop
- ☐ D. Compilation error

4. Which of the following options is correct?

`college.classroom.students=250;`

- A. Structure students is nested within structure college.
- ☒ B. Structure classroom is nested within structure college.
- C. Structure college is nested within structure classroom.
- D. Structure college is nested within structure students.

5. Consider the following code snippet:

If x is an array of integer, then the value of `&x[i]` is same as:

- i. `&x[i-1] + sizeof (int)`
- ii. `x + sizeof (int) * i`
- iii. `x+i`
- iv. All of the above

A. i B. ii ☒ C. iii D. iv

6. In the following C code, in which order would the functions be called?

`x = (f1(23,14) * f2 (12/14)) / f3();`

- ☒ A. f1, f2, f3
- B. f3, f2, f1
- C. f3, f1, f2
- D. Compiler dependent

7. Which of the following statements should be used to print the value pointed to by the generic pointer 'p'?

`int i = 1000;`

`void *p = &i;`

- A. `printf("%d", *p);`
- B. `printf("%d", *(int)p);`
- C. `printf("%d", *(int*)p);`
- ☒ D. `printf("%d", (int*)*p);`

8. If max is a function that returns the larger of the two integers, given as arguments, then which of the following statements finds the largest of three given numbers.

- i. `max(max(a,b),max (a,c))`
- ii. `max(max(a,b),max (b,c))`
- iii. `max(b,max (a,c))`
- iv. All of these

A. i B. iii C. i, iii and iv ☒ D. iv

9. Under which of the following conditions, the size of a 1D array need not be specified in the array syntax?

- i. when initialization is a part of declaration
- ii. when it is a declaration only
- iii. when it is a formal parameter and an actual argument
- iv. All of the above

A. i & ii B. ii & iii ☒ C. i & iii D. iv

10. `int y[2][3]={1,2,3},{4,5,6};`

`int (*ptr)[3]=y;`

`printf("%p",ptr+1);`

Which of the following address would be printed if the integer is 4 bytes long?

- A. `y+1byte`
- B. `y+4bytes`
- C. `y+8bytes`
- ☒ D. `y+12bytes`

11. Consider the function fun defined below.

```

struct mango {
    int value;

    struct mango * next;};

int fun(struct mango *p) {

    return ( (p == NULL) || (p -> next == NULL) || ((p -> value <= p -> next -> value) && fun(p->next))); }

```

For a given linked list p, the function fun returns 1 if and only if

- ☒ A. Not all elements in the list have the same data value
 - ☐ B. The elements in the list are sorted in non-decreasing order of value
 - ☐ C. The elements in the list are sorted in non-increasing order of value
 - ☐ D. None of them
12. Compared to an array, which of the following statements regarding the Linked List data structure is true?
- ☐ A. Unlike arrays, nodes of the linked list cannot be created on the stack.
 - ☐ B. Typically, Linked Lists implementations prohibit random access.
 - ☐ C. While linked lists can have their size changed at any moment, the size of an array must be predetermined.
 - ☒ D. All of these

13. How many times will printf execute in the following code:

```

int main() {
    void display();
    display(-4);}

```

```

void display(int num) {
    printf("C programming");
    if (num++ == 0) return;
    display(num);
    num++; }

```

- A. 4 B. 5 C. 3 ☒ D. Compilation error

14. Fill the question mark to print **C Program** as output.

```

char ptr[] = "C Program";
void *vptr;
vptr = ptr;
printf("%s", ?);

```

- A. *(char **)vptr B. (char)vptr C. *(char *)vptr ☒ D. (char *)vptr

15. // insert code here

```

strcat(str1, str2);
puts(str1);

```

Which of the following could be inserted in the above code?

- I. char str1[20] = "Today is ", str2[] = "C Quiz";
- II. char str1[20] = "Today is ", *str2 = "C Quiz";
- III. char *str1 = "Today is ", *str2 = "C Quiz";

Choose one among the following.

- ☒ A. Only I B. Either I or II C. Either I or III D. Either I or II or III

(16-30) What will be the outputs of the following codes? Assume that the necessary standard libraries, and the wrapper (main) functions already exist.

16. <code>char *q = 0;</code> <code>*q = 'A';</code> <code>printf("%c\n", *q);</code> A: 'A' B: 0 C: Run time error D: Compile time error	17. <code>int x = 0, y = 0;</code> <code>if (++y 2 && ++x)</code> <code>printf("%d %d", x, y);</code> A: 1 1 B: 1 0 C: 0 1 D: 0 0	18. <code>char str[] = "BROAD2023", *ptr=str;</code> <code>ptr++;</code> <code>while(*ptr!='\0')</code> <code>printf("%c ", *ptr++);</code> A: BROAD2023 B: Run time error C: Compiler error D: ROAD2023
19. <code>char* str="Grade=BI";</code> <code>str[6] = 'A';</code> <code>printf("%s", str);</code> A: Grade=BI B: Grade=AI C: Run time error D: Compiler error	20. <code>const int a = 15;</code> <code>int * const ptr = &a;</code> <code>++ptr;</code> <code>printf("%d ", *ptr);</code> A: 15 B: 16 C: Garbage D: Error	21. <code>char *c[] = {"C Program", "MCQ", "COMPARE", "TEST"};</code> <code>char **cp[] = {c+3, c+2, c+1, c};</code> <code>char ***cpp = cp;</code> <code>printf("%s ", **++cpp);</code> <code>printf("%s ", ***cpp);</code> A: COMPARE Program B: COMPARE MCQ C: MCQ COMPARE D: MCQ Program
22. <code>int a[][3] = {0, 1, 2, 3, 4, 5};</code> <code>int (*ptr)[3] = a;</code> <code>++ptr;</code> <code>printf("%d %d\n", (*ptr)[0], (*ptr)[1]);</code> A: 2 3 B: 0 1 C: 3 4 D: 1 2 E: Error	23. <code>void mysteryFunction(int *a, int *b) {</code> <code>*a = *a ^ *b;</code> <code>*b = *a ^ *b;</code> <code>*a = *a ^ *b; }</code> <code>void main() {</code> <code>int x = 5, y = 10;</code> <code>mysteryFunction(&x, &y);</code> <code>printf("%d,%d\n", x, y); }</code> A: 5,10 B: 10,5 C: 5,5 D: 10,10	24. Assume int is 4 bytes long. <pre> struct s { int marks; union test { int x; char arr[10]; int y; } az; } as; int main() { struct s s1; printf("%ld\n", sizeof(s1)); } </pre> 24 B: 14 C: 16 D: None
25. <code>int a[10] = {2,2,4,4,4};</code> <code>int *ptr = a;</code> <code>printf("%u %u", ptr[5], *ptr+6);</code> A: 4 0 B: 4 8 C: 0 8 D: Error	26. Assume int is 2 bytes and char is 1 byte long <pre> char arr[15] = "pointer array"; int *ptr; ptr = arr; printf("%c", ptr[1]); </pre> A: Garbage B: t C: Compile time Error D: o E: t	27. <code>for (int i=0; i<5; i++) {</code> <code>static int c = 0;</code> <code>c++;</code> <code>printf("%d ", c);</code> <code>}</code> A: 1 1 1 1 1 B: 0 1 2 3 4 C: 1 2 3 4 5 D: 0 0 0 0 0 E: Error
28. <code>#define fun1(x) 50</code> <code>#define fun2(x)</code> <code>(fun1(x)<0)?fun1(x)-5:fun1(x)+5</code> <code>void main()</code> <code>{ int x=5;</code> <code>printf("%d", fun2(++x+3)); }</code> A: 4 B: 14 C: 45 D: 55	29. <code>int a[3][3] = {{11,22,33}, {15,25,35}, {50,60,70}, {55,66,77}};</code> <code>printf("%d \n ", *((a+2)+1)-*((a+1)+2)-5);</code> A: Garbage B: 30 C: 7 D: 20	30. <code>char a[7] = "PILANI";</code> <code>int m=1;</code> <code>for(; a[m]!='\0'; m++)</code> <code>if((m%2)!=0)</code> <code>printf("%c", a[++m]);</code> A: Error B: IAI C: PLN D: LN

Part B

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ID No: _____ Name: _____

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Invigilator's Signature:

- This is an in-built question paper. Write the answers only in the space provided.
- Don't let your answers flow outside the boxes.
- **The marking is strictly binary. Each blank will be awarded marks only if fully correct.**
- **Over-written answers of any kind will not be accepted for rechecks.**
- Assume that the necessary standard libraries, and the wrapper (main) functions exist wherever required.

Recheck requests (write in bullets for each question)	Remarks

1. Your task is to complete the function ***printPascalsTriangle*** to print Pascal's Triangle up to the specified number of rows ***n*** in the code below. Pascal's Triangle is a mathematical construct where each number (coefficient) is the sum of the two numbers directly above it in the previous row. The triangle begins with the number 1 at the top.

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1

[3+3+2 =8M]

```
void printPascalsTriangle(int n) {
    for (int i = 0; i < n; i++) {
        int coefficient = 1;

        for (_____ ) { //Add leading spaces for alignment
            printf(" ");
        }

        for (_____ ) { //Print the numbers (coefficients)
            printf("%d ", coefficient);

            _____ //Update the coefficient for the next term
        }
        printf("\n");
    }
}
```

2. In a database of football players, a structure is defined with the name, jersey number, country, number of goals in NUM_MATCHES matches of a tournament and total goals of a player. Based on the total number of goals, the following incomplete program displays the details of the highest goal scorer. Fill in the blanks accordingly.

[7+7+7 = 21M]

```
#define NUM_MATCHES 3
```

```
typedef struct {  
    char name[20];  
    int jersy_no;  
    int goals[NUM_MATCHES];  
    char country[20];  
    int total_goals;  
} Player;
```

```
void Score(Player p[], int n) { //Function to calculate the total goals of each player  
    for(int k=0;k<n;k++)
```

```
{  
    _____  
    _____  
    _____  
}
```

```
}
```

```
void readInfo(Player p[], int n) { //Function to read each player's info  
    for(int i=0;i<n;i++)
```

```
{  
    printf("Enter the name, jersy_no, and country name of each player\n");
```

```
    _____  
    printf("Enter the goal records for each player\n");
```

```
    _____
```

```
    }  
    Score(p,n);
```

```
}
```

```
int maxGoals(Player p[], int n) { //Function to return the index (pos) of the player with maximum goals  
    int max=p[0].total_goals, pos=0;
```

```
    for(_____){
```

```
        _____
```

```
        _____
```

```

    }
    return pos;
}

int main() {
    int num_players;
    scanf("%d", &num_players);
    Player p[num_players];
    readInfo(p, num_players);
    int k=maxGoals(p, num_players);
    printf("Details of the highest Goal Scorer\n");
    printf("Name: %s \t jersy_no:%d \t Country:%s\t Total Goals:%d\n ", p[k].name, p[k].jersy_no, p[k].country,
p[k].total_goals);
    return 0;
}

```

3. Complete the following incomplete program to concatenate two strings using pointers. **[3+3 = 6M]**

```

#define MAX_SIZE 100 // Maximum string size
int main() {
    char str_A[MAX_SIZE], str_B[MAX_SIZE];
    char * str1 = str_A;
    char * str2 = str_B;

    /* Input two strings from user */
    printf("Enter first string: ");
    gets(str_A);
    printf("Enter second string: ");
    gets(str_B);

    while(_____); //Move till the end of str_A

    while(_____); //Copy str_B to str_A
    printf("Concatenated string = %s", str_A);
    return 0;
}

```

4. The following C function **printMiddle** prints the middle element of a linked list using the concept of slow and fast pointers. The slow pointer moves one node at a time, while the fast pointer moves two nodes at a time. Complete the function accordingly. **[2*3 = 6M]**

```

struct Node {
    int data;
    struct Node* next;
};

```

// Function to get the middle of the linked list. For even number of nodes, return the first middle node

```
void printMiddle(struct Node *head) {
    struct Node *slow_ptr = head;
    struct Node *fast_ptr = head;

    if (head!=NULL) {
        while (_____) {
            _____
            _____
        }
        printf("The middle element is %d \n", slow_ptr->data);
    }
}
```

5. Fill in the blanks for the program given below: **[2+2=4M]**

```
int *alloc_mem_and_set_value_as_4() {
    int *p;
    _____; // Allocate memory dynamically for p
    _____; // Set the value of the location allocated to 4
    return p;
}
```

6. Total how many int elements are present in the following dynamic 2D array? _____ **[2M]**

```
int **arr2d = (int **)malloc(3 * sizeof(int *));
for (int i = 0; i < 3; i++)
    arr2d[i] = (int *)malloc((i/2 + i + 3) * sizeof(int));
```

7. For each of the following ((a) to (e)), choose the most appropriate declarations from the options given below in 1 to 12. **Just write the option from 1 to 12, no need to write the whole declaration.** **[5*2 = 10M]**

- a) An entity that can contain the address of a float variable. _____
 b) An entity which points to a float array of hundred values. _____
 c) An entity that can contain the addresses of hundred float variables. _____
 d) An entity that can store a scalar value at every unit cubic inch in a space whose dimensions are 10 inches long, 30 inches wide and 5 inches tall. _____
 e) An entity that can store the 3d positional coordinates at unit length in three dimensions along with a scalar value of a space which is 10 inches long, 30 inches wide and 5 inches tall.

1. float *p;	2. float **p;	3. ELEM *p;	4. ELEM **p;
5. float p[100];	6. float *p[100];	7. float (*p)[100];	8. float p[10][30][5];
9. ELEM p[10][30][5];	10. ELEM p[100];	11. ELEM *p[100];	12. ELEM (*p)[100];

```
typedef struct
{
    int length;
    int breadth;
    int height;
    float val;
} ELEM;
```
